Employee Attrition Prediction Analysis

A Predictive Modeling Approach to Workforce Retention

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Abstract

Employee attrition is a major concern for organizations as it leads to increased recruitment costs and loss of experienced workforce. This project aims to predict whether an employee is likely to leave a company using machine learning techniques.

The system uses HR analytics data and applies classification algorithms such as Logistic Regression, Decision Tree, and Random Forest. The performance of these models is compared to identify the most accurate model for prediction.

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1. Problem Statement

Employee turnover, also known as attrition, refers to the situation where employees leave an organization for various reasons. High attrition rates create significant financial and operational challenges due to recruitment costs, training expenses, and the loss of valuable institutional knowledge.

The main objective of this project is to analyze HR metrics to predict which employees are likely to leave the organization. By identifying patterns and trends in employee data, management can take proactive steps to implement effective retention strategies. This helps in reducing turnover, improving workforce stability, and increasing overall organizational productivity.

2. Dataset Description

The study uses a comprehensive HR dataset consisting of 1,470 employee records. Each record corresponds to an individual employee and includes 35 distinct attributes covering demographic details, job-related information, and satisfaction levels.

The features are grouped into the following categories:

Demographics: Age, Gender, MaritalStatus
This includes basic personal information about each employee.

Employment Details: Department, JobRole, TotalWorkingYears
These attributes describe the employee's professional background and current role within the organization.

Compensation: MonthlyIncome, PercentSalaryHike, StockOptionLevel
This group captures the financial benefits and incentives provided to employees.

Work-Life Factors: OverTime, WorkLifeBalance, DistanceFromHome
These variables reflect aspects that influence the employee's daily work-life balance.

Target Variable: Attrition
A binary indicator (Yes/No) showing whether the employee has left the organization.

Data cleaning confirmed that there are no missing values in any of the columns, ensuring a high-quality dataset suitable for predictive modeling.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the relationships between different features and employee attrition. The analysis revealed several important insights:

Class Imbalance:
The dataset is imbalanced, with a much larger proportion of employees staying compared to those leaving. This imbalance was taken into account during model evaluation.

Impact of Overtime:
Employees who work overtime were found to have a significantly higher likelihood of leaving the organization compared to those who do not.

Income Levels:
Lower monthly income showed a strong correlation with employee attrition, indicating that financial dissatisfaction is a key factor influencing employee turnover.

4. Methodology: Random Forest Classification

For predictive modeling, the Random Forest algorithm was used. It is an ensemble learning technique that constructs multiple decision trees and combines their outputs to produce more accurate and stable predictions.

4.1 Data Preprocessing

Categorical variables such as *BusinessTravel*, *Department*, and *EducationField* were converted into numerical format using Label Encoding, making the data suitable for machine learning algorithms.

After preprocessing, the dataset was split into training and testing sets in an 80:20 ratio.

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier()
model.fit(X_train, y_train)
```

5. Results and Model Evaluation

The model's performance was evaluated on the hold-out test set using accuracy, precision, recall, and F1-score metrics.

Evaluation Results:

- **Overall Accuracy:** 87.41%
- **Precision (Class 0 – No Attrition):** 0.88
- **Recall (Class 0 – No Attrition):** 1.00
- **F1-Score (Class 0 – No Attrition):** 0.93

Although the model performs well in identifying employees who are likely to stay, it shows limited effectiveness in detecting employees who are likely to leave (Class 1). The recall for Class 1 is low (0.08), which is a common issue in imbalanced datasets where the majority class dominates the predictions.

6. Feature Importance Analysis

The Random Forest model identified the most important features influencing employee attrition. Analyzing feature importance helps HR teams focus on key areas for improving retention strategies.

Monthly Income:
This was found to be one of the most significant factors affecting attrition, indicating that compensation plays a major role in employee retention.

Age:
Younger employees were more likely to leave the organization, making age an important predictor in the model.

Daily Rate & Total Working Years:
Both professional experience and daily earnings also had a notable impact on the model's predictions, showing that career progression and compensation structure influence attrition behavior.

7. Conclusion & Recommendations

The analysis demonstrates that machine learning can effectively identify employees at risk of attrition with high overall accuracy. Based on the insights derived from the model, the following organizational strategies are recommended:

1. Compensation Review:
Assess and adjust salary structures for roles with higher attrition rates to ensure competitive and fair compensation.

2. Overtime Management:
Monitor and reduce excessive overtime, as it has been identified as a key factor contributing to employee turnover.

3. Career Development Opportunities:
Focus on improving growth prospects by addressing factors such as *YearsInCurrentRole* and *YearsSinceLastPromotion*, ensuring employees have clear career progression pathways.

8. References

IBM HR Analytics Employee Attrition & Performance Dataset, available on Kaggle.

Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, 12, pp. 2825–2830, 2011.

Jake VanderPlas, *Python Data Science Handbook*, O'Reilly Media, 2016.